

VIMAL SHARMA

Analytical Modelling Manager

vimal2017sharma@gmail.com | 9205714199 | Bangalore | [LinkedIn](#) | [GitHub](#)

KEY SKILLS

Python · PySpark · SQL · Scala · Apache Airflow · GCP | AML · Transaction Monitoring · Financial Crime | Prompt Engineering · GenAI · LLMs · Agentic AI | Quantexa · Graph Analytics · Entity Resolution · Model Development | AI Nerd

PROFESSIONAL SUMMARY

Analytical Modelling Manager with 3+ years at HSBC in AML and financial crime risk. Expert in PySpark, Python, and Agentic AI to build anomaly detection, network-based transaction monitoring, and GenAI-powered investigation tools at scale. Proven track record of \$240K+ in savings, 1,000+ hours automated, and 30% alert reduction. Known for bridging data engineering and product thinking to drive compliance decisions across global markets.

WORK EXPERIENCE

Manager, Analytical Modelling – Decision Science

May 2025 – Present

HSBC, Bangalore

- Saved \$240K annually and reduced alerts by 26% by deploying rule-based and behavioural anomaly detection models using PySpark and Scala.
- Designed and tuned 15+ rule-based and statistical models for AML transaction monitoring, setting risk-based thresholds leading to 20% increase in investigator efficiency.
- Automated data quality monitoring (DUSE) across 4 markets, improving data quality by 40% and cutting cycle time from 3 months to 1 week.
- Built GenAI prompt workflows for investigators to auto-close cases with embedded investigation summaries, deployed in production.
- Improved entity resolution by 80% using Quantexa with auto coldlisting thresholds.
- Built ETL pipelines using Apache Airflow and PySpark to evaluate alert impact post-threshold tuning.
- Using Network Graph mapping relationships between entities (people, accounts, companies) to detect hidden connections.
- Led proof-of-concept (POC) evaluations of 3 Gen AI solutions for financial crime use cases, presenting findings to senior leadership

Assistant Manager – Decision Science

Jun 2023 – Apr 2025

HSBC, Bangalore

- Deployed Quantexa network-based transaction monitoring across 5+ HSBC markets.
- Built AML customer segmentation models and assessed banking infrastructure for compliance risk gaps.
- Authored model documentation and business requirement documents (BRDs) reviewed and approved by model validation teams and senior leadership, reducing review cycle time by 25%
- Led 2-week sprint cycles across a cross-functional team of 8 (data engineers, IT, analysts, and compliance SMEs), delivering 4 transaction monitoring models within a 6-month roadmap
- Facilitated daily standups, sprint reviews, and retrospectives, improving team velocity by 20% over 3 quarters and maintaining a sprint completion rate of 90%+.

PROJECTS

TxnGuard – Agentic AML Transaction Monitor [GitHub Link](#)

- Built an agentic AML investigation system using LLMs to analyse transaction patterns, enrich alerts with PEP/sanctions/offshore risk signals, and generate explainable SAR risk scores.

Automated Volume-Based Order Execution [GitHub Link](#)

- Automated trading system using Fyers API v3 with real-time volume analysis, buy/sell triggers, backtesting, and portfolio risk controls.

EDUCATION

MBA, Finance - 2023

Christ University, Bangalore | GPA : 3.6 / 4

BA (Hons), Economics - 2020

Amity University, Noida | CGPA : 7.75 / 10

CERTIFICATIONS

CFA Level 1 (CFA Institute) · International Compliance Association · ISO 20022 Messaging Standards · Bloomberg Terminal · Lean Six Sigma Yellow Belt · Design Thinking · AI Certifications